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Studierichting: Informatica

Oefeningenreeks 1C nr. 46.2

Opgave: $z^4 = -1$, waarbij $r = 1$ en $\theta = \pi$

$$w = \sqrt[4]{1} \cdot \text{cis} \left(\frac{\pi + k \cdot 2\pi}{4} \right) \quad k \in \{0, 1, 2, 3\}$$

$$\begin{array}{llll} w_0 &= \text{cis} \frac{\pi}{4} &= \frac{\sqrt{2}}{2} + i \frac{\sqrt{2}}{2} & \text{voor } k = 0 \\ w_1 &= \text{cis} \frac{3\pi}{4} &= -\frac{\sqrt{2}}{2} + i \frac{\sqrt{2}}{2} & \text{voor } k = 1 \\ w_2 &= \text{cis} \frac{5\pi}{4} &= -\frac{\sqrt{2}}{2} - i \frac{\sqrt{2}}{2} & \text{voor } k = 2 \\ w_3 &= \text{cis} \frac{7\pi}{4} &= \frac{\sqrt{2}}{2} - i \frac{\sqrt{2}}{2} & \text{voor } k = 3 \end{array}$$

References